

PHASE 1 – Azure Fundamentals (Foundation)

Goal:

Understand core Azure concepts and how cloud works.

Cloud Basics

- What is IaaS, PaaS, SaaS?
- Public vs Private vs Hybrid cloud
- Regions & Availability Zones
- Resource Groups

Example:

- Create a **Resource Group**
 - Deploy a simple **Azure App Service**
 - Deploy your existing ASP.NET Core Web API
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Core Azure Services You Must Know

Azure App Service

Used to host:

- ASP.NET Core Web APIs
- Angular apps (static hosting)

Example:

- Deploy your ASP.NET Core API to App Service
 - Publish Angular app to App Service
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Azure SQL Database

Managed SQL Server in cloud.

Example:

- Create Azure SQL DB
 - Connect it from ASP.NET Core using connection string
 - Use EF Core migration
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Azure Roadmap for .NET + Angular Developer

Azure Storage Account

- Blob Storage (file storage)
- Queue Storage
- Table Storage

Example:

- Upload profile images from Angular
 - Store in Blob Storage
 - Save URL in SQL DB
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Azure Active Directory (Azure AD / Entra ID)

Very important for enterprise apps.

Example:

- Secure ASP.NET Core Web API using Azure AD
 - Angular login using MSAL
 - Role-based authorization
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PHASE 2 – Real-World App Architecture

Now we move from basics to professional architecture.

Azure App Configuration & Key Vault

Azure Key Vault

Store:

- DB connection strings
- API keys
- Secrets

Example:

- Store DB password in Key Vault
 - Access it in ASP.NET Core
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Azure Roadmap for .NET + Angular Developer

Azure Application Insights

Monitoring + Logging.

 Example:

- Track API failures
 - Track Angular frontend errors
 - Monitor performance
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Azure Service Bus (Important for scalable apps)

Used for:

- Microservices communication
- Background processing

 Example:

- Order placed → Send message to Service Bus
 - Background worker processes payment
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PHASE 3 – DevOps & Deployment (Very Important)

As a developer, this will boost your salary 

Azure DevOps

Using:

- Repos
- Pipelines
- CI/CD

 Example:

- Push code to Azure DevOps repo
 - Build ASP.NET Core
 - Build Angular
 - Deploy automatically to Azure App Service
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7 GitHub Actions (Alternative to DevOps)

If using GitHub:

- Create CI/CD pipeline
 - Auto-deploy on merge to main branch
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● PHASE 4 – Containers & Modern Architecture

8 Docker + Azure Container Registry

👉 Example:

- Dockerize ASP.NET Core API
 - Push image to Azure Container Registry
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9 Azure Kubernetes Service (AKS)

For microservices & scaling.

👉 Example:

- Deploy API container to AKS
 - Configure scaling
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● PHASE 5 – Advanced / Enterprise Level

10 Azure Functions (Serverless)

👉 Example:

- Create HTTP-triggered function
 - Background email sender
 - Scheduled cleanup job
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11 Azure API Management

Used in enterprise apps.

👉 Example:

- Expose multiple microservices under one API gateway

Azure Roadmap for .NET + Angular Developer

- Add rate limiting
 - Add authentication
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Suggested Learning Order (Very Practical)

Follow this order strictly:

1. Azure Fundamentals
 2. App Service
 3. Azure SQL
 4. Blob Storage
 5. Azure AD
 6. Key Vault
 7. Application Insights
 8. Azure DevOps CI/CD
 9. Docker
 10. Azure Functions
 11. AKS
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Real Practice Project (Best Way to Learn)

Build this project step-by-step:

Project: E-Commerce Mini App

Stack:

- ASP.NET Core Web API
- Angular frontend
- Azure SQL
- Blob Storage
- Azure AD auth
- Service Bus for orders
- App Insights monitoring
- CI/CD pipeline

Deploy fully on Azure.

That single project will teach you 70% of real-world Azure.

Certification Path (Optional but Valuable)

Start with:

-  **AZ-900 (Azure Fundamentals)**

Then:

-  **AZ-204 (Azure Developer Associate)**

These align perfectly with your background.

Suggested Timeline

If you study 1–2 hours daily:

- Month 1 → Fundamentals + App Service + SQL
 - Month 2 → Security + Storage + Monitoring
 - Month 3 → DevOps + Docker
 - Month 4 → Advanced services
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Final Advice (From Experience)

As a .NET developer:

 Focus more on:

- App Service
- Azure SQL
- Azure AD
- DevOps
- Key Vault
- Application Insights

These are used in 80% of enterprise jobs.

Azure Roadmap for .NET + Angular Developer



WEEK 1 – Azure + Git Foundations



Goal:

Understand Azure basics + Git workflow.



Day 1 – Cloud & Azure Basics

Learn:

- What is Cloud?
- IaaS vs PaaS vs SaaS
- Regions
- Resource Groups

Hands-on:

- Create Azure Free Account
 - Create Resource Group
 - Explore Azure Portal
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Day 2 – Git Basics (Very Important)

Learn:

- git init
- git clone
- git add
- git commit
- git push
- git pull
- Branches

Hands-on:

- Create GitHub repository
- Push your ASP.NET Core API project

Platform:

Use GitHub

Day 3 – Branching & Pull Requests

Learn:

- Feature branch workflow
- Pull Request (PR)
- Code review
- Merge conflicts

Practice:

1. Create branch: feature/login
 2. Make changes
 3. Push branch
 4. Create Pull Request
 5. Merge to main
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Day 4 – Azure App Service (Hosting API)

Learn:

- What is App Service?
- App Service Plan
- Pricing tiers

Hands-on:

- Create Web App
- Deploy ASP.NET Core API manually from Visual Studio

Service:

Azure App Service

Day 5 – Deploy Angular to Azure

Options:

- App Service
- Static Web Apps

Learn:

- Build Angular production build
- Deploy to Azure

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Service:

Azure Static Web Apps

Day 6 – Azure SQL Database

Learn:

- Create Azure SQL
- Firewall rules
- Connection strings

Hands-on:

- Connect ASP.NET Core API to Azure SQL
- Run EF Core migration

Service:

Azure SQL Database

Day 7 – Mini Project v1

Create:

- Simple Product API
 - Angular UI
 - Store data in Azure SQL
 - Deploy both to Azure
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WEEK 2 – CI/CD & Automation

Goal:

Automate build & deployment.

Day 8 – Introduction to CI/CD

Understand:

- What is CI?
- What is CD?
- Why pipelines?

Tool options:

Azure Roadmap for .NET + Angular Developer

- GitHub Actions
- Azure DevOps

(For you → start with GitHub Actions, easier)

Day 9 – CI Pipeline for ASP.NET Core

Create workflow:

- Trigger on push to main
- Build .NET project
- Run tests

Example steps:

- Setup .NET
 - Restore
 - Build
 - Test
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Day 10 – CD Pipeline (Deploy API)

Extend workflow:

- After successful build
- Deploy to Azure App Service automatically

Now:

Every push → auto deployment 

Day 11 – Angular CI/CD

Create pipeline:

- Install Node
 - npm install
 - npm build
 - Deploy to Azure
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Day 12 – Environments

Learn:

- Dev
- QA
- Prod

Create:

- Separate Azure App Service for Dev & Prod
 - Deploy using branch strategy:
 - develop → Dev environment
 - main → Production
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Day 13 – Secrets & Variables

Learn:

- Store secrets in GitHub
- Avoid hardcoded passwords

Use:

- GitHub Secrets
 - Azure App Settings
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Day 14 – Mini Project v2 (Fully Automated)

Now your project should:

- ✓ Auto-build
 - ✓ Auto-test
 - ✓ Auto-deploy
 - ✓ Use branch-based deployment
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WEEK 3 – Security, Monitoring & Storage

Day 15 – Azure Blob Storage

Service:

Azure Blob Storage

Practice:

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- Upload images
 - Store URL in DB
 - Retrieve in Angular
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Day 16 – Azure Key Vault

Service:

Azure Key Vault

Practice:

- Store DB connection string
 - Access in ASP.NET Core
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Day 17 – Authentication with Azure AD

Service:

Microsoft Entra ID

Practice:

- Secure API
 - Login from Angular
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Day 18 – Application Insights

Service:

Azure Application Insights

Practice:

- Monitor API
 - Track exceptions
 - View performance
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Day 19 – Logging & Diagnostics

Learn:

- Log levels
 - Structured logging
 - Viewing logs in Azure Portal
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Day 20–21 – Improve Project

Add:

- Authentication
- File upload
- Logging
- Monitoring

Now your project looks enterprise-level.

WEEK 4 – DevOps Deep Understanding

Day 22 – Azure DevOps Overview

Platform:

Azure DevOps

Learn:

- Boards
 - Repos
 - Pipelines
 - Artifacts
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Day 23 – Create Pipeline in Azure DevOps

- Create repo
 - Push code
 - Build pipeline
 - Release pipeline
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Day 24 – Pull Request Policies

Learn:

- Required reviewers
- Build validation
- Branch protection rules

Azure Roadmap for .NET + Angular Developer

Implement in:
GitHub or Azure DevOps

Day 25 – Docker Basics

Learn:

- What is Docker?
- Dockerfile
- Build image
- Run container

Dockerize your ASP.NET Core API.

Day 26 – Azure Container Registry

Service:

Azure Container Registry

Push Docker image to ACR.

Day 27 – Deploy Container to Azure

Deploy container to:

- App Service (container option)
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Day 28 – Cost Management

Learn:

- Azure Cost Management
- Budgets
- Alerts

Avoid surprise bills 💰

Day 29 – Full Revision

Revise:

- Git workflow
- CI/CD

Azure Roadmap for .NET + Angular Developer

- Azure services
- Security

Day 30 – Final Project Deployment

Deploy fully working system:

- ✓ API
- ✓ Angular
- ✓ SQL
- ✓ Blob Storage
- ✓ Auth
- ✓ Monitoring
- ✓ CI/CD